DSP · MOCK

DSP mock B

Answers and worked solutions · Day 24 · Mon 16 Nov · pack DX2

Marking guide

Mark your script against these solutions. Give part marks for correct method even when the arithmetic slips. Every number below has been checked in Python (numpy/scipy/sympy). The “Reread” line names the study pack to go back to if you lost marks.

Question 1: Correlation and delay estimation [5]

(a) [1] x is nonzero for $n = 0, 1, 2$.

- $r_{xx}(0) = 1^2 + 2^2 + (-1)^2 = 6$
- $r_{xx}(1) = x(1)x(0) + x(2)x(1) = 2 - 2 = 0$
- $r_{xx}(2) = x(2)x(0) = -1$

Autocorrelation is even, so

$$r_{xx}(l) = \{-1, 0, \underset{\uparrow}{6}, 0, -1\}, \quad l = -2, \ldots, 2.$$

(b) [2] $r_{yx}(l) = \sum_n y(n)x(n-l) = y(l) + 2y(l+1) - y(l+2)$. Slide x along y. (This equals $y(l) * x(-l)$, so you can also do it as a convolution with the flipped pulse.)

l	-2	-1	0	1	2	3	4	5	6
$r_{yx}(l)$	-1	2	2	-4	-1	**12**	-1	0	1

For example, $r_{yx}(3) = y(3) + 2y(4) - y(5) = 2 + 8 + 2 = 12$.

(c) [1] $\hat{d} = \arg\max_l |r_{yx}(l)| = 3$ samples. From $d = 2F_s r/c$:

$$r = \frac{\hat{d}\,c}{2F_s} = \frac{3 \times 3 \times 10^8}{2 \times 10^7} = \mathbf{45\ m}.$$

(d) [1] Near the peak, $r_{yx}(\hat{d}) \approx a\, r_{xx}(0)$, so $\hat{a} = 12/6 = \mathbf{2}$. Range resolution is $c/(2F_s) = 3 \times 10^8/(2 \times 10^7) = \mathbf{15\ m}$ per sample of delay.

MARKS: (a) 1 for all five values. (b) 2 for the full table, or 1 if one or two entries are wrong. (c) ½ for $\hat{d}$, ½ for r. (d) ½ for a, ½ for resolution.

COMMON MISTAKES

- Mixing up the conventions. T3 Q7 uses $R_{xy}[l] = \sum x[k]y[k+l]$, which is the lecture's $r_{yx}(l)$. Whatever the convention, the peak must sit at a **positive** lag equal to the delay. If you get -3, your lag sign is flipped.
- Picking the largest value rather than the largest magnitude. Here it makes no difference, but the course's rule is $\arg\max |r_{yx}|$.
- Forgetting the factor 2 in $d = 2F_s r/c$. The signal makes a round trip.

REREAD: D1 (L03 correlation and radar case study).

Question 2: LCCDE with complex modes [6]

(a) [1] Characteristic equation: $\lambda^2 - \lambda + 0.5 = 0$, so

$$\lambda_{1,2} = 0.5 \pm j0.5 = \tfrac{1}{\sqrt{2}}\, e^{\pm j\pi/4}.$$

The modes are $\left(\tfrac{1}{\sqrt{2}}\right)^n e^{\pm j\pi n/4}$. Both have $|\lambda| = 0.7071 < 1$, so the causal system is **stable**.

(b) [1] $h(n) = \delta(n) + h(n-1) - 0.5h(n-2)$:

$$h(n) = \{\underset{\uparrow}{1}, 1, 0.5, 0, -0.25, -0.25, -0.125\}, \quad n = 0, \ldots, 6.$$

(c) [2] Complex-conjugate modes combine into a real form:

$$h(n) = \left(\tfrac{1}{\sqrt{2}}\right)^n \left[A\cos\tfrac{\pi n}{4} + B\sin\tfrac{\pi n}{4}\right] u(n).$$

- $h(0) = 1 \Rightarrow A = 1$.
- $h(1) = \tfrac{1}{\sqrt{2}}\left(\tfrac{1}{\sqrt{2}} + \tfrac{B}{\sqrt{2}}\right) = \tfrac{1+B}{2} = 1 \Rightarrow B = 1$.

$$\boxed{h(n) = \left(\tfrac{1}{\sqrt{2}}\right)^n \left[\cos\tfrac{\pi n}{4} + \sin\tfrac{\pi n}{4}\right] u(n) = 2^{(1-n)/2} \cos\left(\tfrac{\pi(n-1)}{4}\right) u(n)}$$

Check: $n = 2$ gives $\tfrac{1}{2}(0+1) = 0.5$, and $n = 4$ gives $\tfrac{1}{4}(-1+0) = -0.25$. Both match (b).

(d) [2] $H(\omega) = \dfrac{1}{1 - e^{-j\omega} + 0.5e^{-j2\omega}}$.

- $H(0) = \dfrac{1}{1 - 1 + 0.5} = 2$.
- $H(\pi/2) = \dfrac{1}{1 - (-j) + 0.5(-1)} = \dfrac{1}{0.5 + j} = 0.4 - j0.8 = 0.8944\angle -1.107$ rad (-63.43°).

$$\boxed{y_{ss}(n) = 8 + 1.789\cos\left(\tfrac{\pi}{2}n - 1.107\right)}$$

MARKS: (a) ½ modes, ½ stability with reason. (b) 1. (c) 1 for the real form, 1 for A and B. (d) 1 for each frequency (gain and phase applied correctly).

COMMON MISTAKES

- Writing the real form with the wrong radius, for example $(0.5)^n$ instead of $|\lambda|^n = (1/\sqrt{2})^n$.

- Fitting A and B with $h(-1) = h(-2) = 0$ and getting confused. Use $h(0)$ and $h(1)$ from the recursion.
- Taking $e^{-j\pi/2} = +j$. It is $-j$.
- Doubling the DC term or forgetting to scale it: the DC part is $4H(0) = 8$.

REREAD: D2 (modes and $h(n)$ from an LCCDE), D4 (sinusoidal steady state).

Question 3: DTFT shortcut evaluations [5]

The samples are $x(-2) = 1$, $x(-1) = -2$, $x(0) = 3$, $x(1) = 0$, $x(2) = -1$ and $x(3) = -4$.

(a) [1] $X(0) = \sum_n x(n) = 1 - 2 + 3 + 0 - 1 - 4 = \mathbf{-3}$. A negative real number has $\angle X(0) = \boldsymbol{\pi}$ (or $-\pi$).

(b) [1] $X(\pi) = \sum_n (-1)^n x(n) = 1 + 2 + 3 + 0 - 1 + 4 = \mathbf{9}$. (Signs: $n = -2$ gives $+$, $n = -1$ gives $-$, and so on.)

(c) [1] From the inverse DTFT, $x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$. So the integral is $2\pi x(2) = \mathbf{-2}\pi \approx -6.283$.

(d) [1] Parseval:
$\int_{-\pi}^{\pi} |X(\omega)|^2 d\omega = 2\pi \sum |x(n)|^2 = 2\pi(1 + 4 + 9 + 0 + 1 + 16) = \mathbf{62}\pi \approx 194.8$.

(e) [1] Differentiation in frequency: $n\,x(n) \leftrightarrow j\,dX/d\omega$, so $|dX/d\omega|^2 = |\text{DTFT of } nx(n)|^2$. Apply Parseval to $n\,x(n)$:

$$\int_{-\pi}^{\pi} \left| \frac{dX}{d\omega} \right|^2 d\omega = 2\pi \sum n^2 |x(n)|^2 = 2\pi(4 + 4 + 0 + 0 + 4 + 144) = \mathbf{312}\pi \approx 980.2.$$

MARKS: 1 per part. In (a), give ½ if $X(0)$ is right but the angle is given as 0.

COMMON MISTAKES

- Putting the origin in the wrong place. The arrow is under the 3, so the first sample is at $n = -2$. This flips the signs in $X(\pi)$ and changes (c) and (e).
- Saying $\angle X(0) = 0$ because $X(0)$ is real. Real and negative means $\pm\pi$.
- Forgetting the 2π in Parseval, or using $\frac{1}{2\pi}$ on the wrong side.
- Using $\sum n|x(n)|^2$ instead of $\sum n^2|x(n)|^2$ in (e).

REREAD: D3 (DTFT properties; T4 Q2 is the course version of this question).

Question 4: Deconvolution via the DFT [6]

(a) [1] The linear convolution has length $L + M - 1 = 3 + 2 - 1 = 4$, which is the length of y. With $N \geq 4$, the N-point circular convolution equals the linear one, so $Y(k) = H(k)X(k)$ holds exactly. Then $X(k) = Y(k)/H(k)$ (provided no $H(k) = 0$), and the IDFT returns $x(n)$ with one trailing zero.

(b) [2] With $W_4 = e^{-j\pi/2} = -j$, we have $Y(k) = \sum_n y(n)(-j)^{nk}$.

- $Y(0) = 4 + 0 + 1 + 1 = 6$
- $Y(1) = 4 + 0(-j) + 1(-1) + 1(j) = 3 + j$
- $Y(2) = 4 - 0 + 1 - 1 = 4$
- $Y(3) = \overline{Y(1)} = 3 - j$

$H(k) = 2 + (-j)^k$, which gives $H(k) = \{3,\ 2 - j,\ 1,\ 2 + j\}$.

(c) [2]

- $X(0) = 6/3 = 2$
- $X(1) = \dfrac{3 + j}{2 - j} = \dfrac{(3 + j)(2 + j)}{5} = \dfrac{5 + 5j}{5} = 1 + j$
- $X(2) = 4/1 = 4$
- $X(3) = 1 - j$

IDFT, $x(n) = \frac{1}{4}\sum_k X(k)\, j^{nk}$:

- $x(0) = \frac{1}{4}(2 + 1 + j + 4 + 1 - j) = 2$
- $x(1) = \frac{1}{4}(2 + (1 + j)j - 4 + (1 - j)(-j)) = \frac{1}{4}(2 + j - 1 - 4 - j - 1) = -1$
- $x(2) = \frac{1}{4}(2 - (1 + j) + 4 - (1 - j)) = 1$
- $x(3) = 0$

$$\boxed{x(n) = \{2, \underset{\uparrow}{-1}, 1\}}$$

Check: $\{2, 1\} * \{2, -1, 1\} = \{4, 0, 1, 1\}$. Correct.

(d) [1] For $h = \{1, 1\}$, $H(2) = 1 + (-1) = 0$. The channel has a zero at $z = -1$ ($\omega = \pi$), which is exactly DFT bin $k = 2$. $Y(2) = 0$ whatever the input, so $X(2) = 0/0$ is undefined. The information at $\omega = \pi$ is destroyed, and the inverse $1/(1 + z^{-1})$ has a pole on the unit circle.

MARKS: (a) 1. (b) 1 for $Y(k)$, 1 for $H(k)$. (c) 1 for $X(k)$, 1 for the IDFT. (d) 1.

COMMON MISTAKES

- Using $W_4 = +j$ in the forward DFT. The forward kernel is $e^{-j2\pi nk/N}$.

- Forgetting the $1/N$ in the IDFT.
- Dividing complex numbers without the conjugate. Multiply top and bottom by $\overline{H(k)}$.
- Choosing $N = 3$ "because x has length 3". N must cover the **output** length.

REREAD: D5 (DFT and IDFT by hand), D6 (DFT filtering and T5 Q4 deconvolution).

Question 5: Overlap-save block convolution [6]

(a) [1] $M = 3$, $N = 4$, so $L = N - M + 1 = \mathbf{2}$ new samples per block. Discard the first $M - 1 = \mathbf{2}$ outputs of each block.

(b) [1] Prepend $M - 1 = 2$ zeros. Each block holds the last 2 samples of the previous block followed by 2 new samples. Output length is $6 + 3 - 1 = 8$, so 4 blocks are needed (pad the tail with zeros):

Block	Input $b(n)$, $n = 0..3$
1	$\{0, 0, 1, 2\}$
2	$\{1, 2, 0, -1\}$
3	$\{0, -1, 3, 1\}$
4	$\{3, 1, 0, 0\}$

(c) [3] With $h = \{1, -1, 2, 0\}$, the samples kept involve no wrap-around:

$$y_c(2) = b(2) - b(1) + 2b(0), \qquad y_c(3) = b(3) - b(2) + 2b(1).$$

Block	Full circular output $y_c(0..3)$	Kept $y_c(2), y_c(3)$
1	$\{0, 4, 1, 1\}$	$1, 1$
2	$\{2, -1, 0, 3\}$	$0, 3$
3	$\{5, 1, 4, -4\}$	$4, -4$
4	$\{3, -2, 5, 2\}$	$5, 2$

$$\boxed{y(n) = \{\underset{\uparrow}{1}, 1, 0, 3, 4, -4, 5, 2\}}$$

This equals the direct linear convolution $h * x$.

(d) [1] Block 1: $y_c(0) = b(0) + h(1)b(3) + h(2)b(2) = 0 - 2 + 2 = 0$ and $y_c(1) = b(1) - b(0) + 2b(3) = 4$. The true values are $y(-2) = y(-1) = 0$. The first $M - 1$ circular outputs wrap around and pick up samples from the **end** of the block (time-aliasing), so they are wrong and must be discarded. Here $y_c(0) = 0$ is right only by coincidence.

MARKS: (a) 1. (b) 1 for correct blocks with the 2-sample overlap. (c) 2 for the block outputs (½ each), 1 for the assembled $y(n)$. (d) 1 with a reason.

COMMON MISTAKES

- Confusing the two methods. Overlap-**save** overlaps the **inputs** and discards outputs. Overlap-**add** uses non-overlapping input blocks and adds the overlapping output tails.
- Forgetting the $M - 1$ leading zeros, so the first block has nothing to discard.
- Stopping after 3 blocks and losing the last two output samples (5, 2).
- Keeping the first L outputs instead of the last L.

REREAD: D6 (block convolution, overlap-save and overlap-add).

Question 6: Every possible sequence for each ROC [6]

(a) [2] Cover-up with $w = z^{-1}$. The numerator is $N(w) = 1 + 11w - 11w^2$.

- Pole 0.5 (set $w = 2$): $A = \dfrac{1 + 22 - 44}{(1-4)(1+6)} = \dfrac{-21}{-21} = 1.$
- Pole 2 (set $w = 0.5$): $B = \dfrac{1 + 5.5 - 2.75}{(1-0.25)(1+1.5)} = \dfrac{3.75}{1.875} = 2.$
- Pole -3 (set $w = -\frac{1}{3}$): $C = \dfrac{1 - \frac{11}{3} - \frac{11}{9}}{(1+\frac{1}{6})(1-\frac{2}{3})} = \dfrac{-35/9}{35/18} = -2.$

$$X(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{2}{1 - 2z^{-1}} - \frac{2}{1 + 3z^{-1}}$$

The degree of the numerator is less than that of the denominator, so there is no direct term.

(b) [3] Use the **given** expression, with coefficients 1, 1 and −1. The poles 0.5, 2 and -3 give four ROCs. Rule: a pole **inside** the ROC's inner radius gives a causal term $p^n u(n)$. A pole **outside** gives an anticausal term $-p^n u(-n-1)$.

ROC	$x(n)$	Type
$\lvert z\rvert < 0.5$	$[-(0.5)^n - 2^n + (-3)^n]u(-n-1)$	anticausal
$0.5 < \lvert z\rvert < 2$	$(0.5)^n u(n) + [-2^n + (-3)^n]u(-n-1)$	two-sided
$2 < \lvert z\rvert < 3$	$[(0.5)^n + 2^n]u(n) + (-3)^n u(-n-1)$	two-sided
$\lvert z\rvert > 3$	$[(0.5)^n + 2^n - (-3)^n]u(n)$	causal

(The anticausal term for $-\frac{1}{1+3z^{-1}}$ is $-\left[-(-3)^n u(-n-1)\right] = +(-3)^n u(-n-1)$.)

(c) [1] Only $0.5 < \lvert z\rvert < 2$ contains the unit circle, so only that sequence is absolutely summable and has a DTFT.

- $x(0) = (0.5)^0 = \mathbf{1}$
- $x(-1) = -2^{-1} + (-3)^{-1} = -0.5 - 0.3333 = \mathbf{-0.8333}$

MARKS: (a) ½ per coefficient, ½ for a correct method or form. (b) ½ per ROC sequence and 1 for the ROC list and classification. (c) ½ for the ROC, ½ for both values.

COMMON MISTAKES

- Listing only three ROCs. n distinct pole radii give $n + 1$ annular ROCs.

- Getting the sign of the anticausal term wrong: $\frac{1}{1-pz^{-1}} \leftrightarrow -p^n u(-n-1)$.
- Using $u(-n)$ instead of $u(-n-1)$.
- In (b), carrying on with your own (a) answer instead of the expression given. In an exam, the "assume" value is what earns the marks.
- Saying the causal sequence is stable. It has poles outside the unit circle.

REREAD: D7 (ROC rules), D8 (inverse by partial fractions; T7 Q1 is the two-pole version).

Question 7: Unilateral z-transform with initial conditions [6]

(a) [2] Use $y(n-1) \leftrightarrow z^{-1}Y^+(z) + y(-1)$ and $y(n-2) \leftrightarrow z^{-2}Y^+(z) + z^{-1}y(-1) + y(-2)$:

$$Y^+ - 0.75\left[z^{-1}Y^+ + 2\right] + 0.125\left[z^{-2}Y^+ + 2z^{-1} + 4\right] = \frac{3}{1-z^{-1}}.$$

The initial-condition terms are $-1.5 + 0.25z^{-1} + 0.5 = -1 + 0.25z^{-1}$. Move them to the right-hand side:

$$Y^+(z)\left(1 - 0.75z^{-1} + 0.125z^{-2}\right) = \frac{3}{1-z^{-1}} + 1 - 0.25z^{-1}.$$

The denominator factors as $(1-0.5z^{-1})(1-0.25z^{-1})$, so

$$Y^+(z) = \underbrace{\frac{3}{(1-z^{-1})(1-0.5z^{-1})(1-0.25z^{-1})}}_{\text{zero-state}} + \underbrace{\frac{1-0.25z^{-1}}{(1-0.5z^{-1})(1-0.25z^{-1})}}_{\text{zero-input}}.$$

(b) [3] The zero-input term cancels to $\dfrac{1}{1-0.5z^{-1}}$.

Zero-state partial fractions (cover-up with $w = z^{-1}$):

- Pole 1 ($w=1$): $\dfrac{3}{(0.5)(0.75)} = 8$
- Pole 0.5 ($w=2$): $\dfrac{3}{(-1)(0.5)} = -6$
- Pole 0.25 ($w=4$): $\dfrac{3}{(-3)(-1)} = 1$

$$Y^+(z) = \frac{8}{1-z^{-1}} - \frac{6}{1-0.5z^{-1}} + \frac{1}{1-0.25z^{-1}} + \frac{1}{1-0.5z^{-1}}$$

$$\boxed{y(n) = [8 - 5(0.5)^n + (0.25)^n]u(n)}$$

(c) [1]

- Zero-input: $(0.5)^n u(n)$. These initial conditions excite only the 0.5 mode.
- Zero-state: $[8 - 6(0.5)^n + (0.25)^n]u(n)$.

Recursion check: $y(0) = 3 + 0.75(2) - 0.125(4) = 4$, and the formula gives $8 - 5 + 1 = 4$. Next, $y(1) = 3 + 0.75(4) - 0.125(2) = 5.75$, and the formula gives $8 - 2.5 + 0.25 = 5.75$. Both match. As $n \to \infty$, $y \to 8 = 3H(1) = 3/0.375$.

MARKS: (a) 1 for the shift property with correct IC terms, 1 for $Y^+(z)$. (b) 1 for the partial fractions, 1 for combining, 1 for $y(n)$. (c) ½ for the split, ½ for the checks.

COMMON MISTAKES

- Sign errors in the IC terms. With $-0.75y(n-1)$ in the equation, the IC contribution is $-0.75y(-1)$, which becomes $+1.5$ on the right-hand side.
- Dropping the $z^{-1}y(-1)$ term from the $y(n-2)$ transform.
- Using $y(-1), y(-2)$ as if they were $y(0), y(1)$.
- Missing the cancellation in the zero-input term. It is not an error to keep it, but the partial fractions get longer.

REREAD: D9 (unilateral z-transform; the L08 slide 14 example is the course version).

Question 8: Digital resonator and allpass filter [6]

(a) [1] $\omega_0 = 2\pi(1000)/8000 = \pi/4$. The denominator is $1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2} = 1 - 1.2728z^{-1} + 0.81z^{-2}$:

$$H(z) = \frac{b_0(1 - z^{-2})}{1 - 1.2728z^{-1} + 0.81z^{-2}}.$$

The zeros at $z = 1$ and $z = -1$ reject **0 Hz (DC)** and **4 kHz ($F_s/2$)** completely.

(b) [2] Evaluate at $e^{j\omega_0}$ with the factored form:

$$|H(\omega_0)| = b_0 \frac{|1 - e^{-j2\omega_0}|}{|1 - re^{j\omega_0}e^{-j\omega_0}|\,|1 - re^{-j\omega_0}e^{-j\omega_0}|} = b_0 \frac{2\sin\omega_0}{(1-r)\,|1 - re^{-j\pi/2}|}.$$

This gives $b_0 \dfrac{1.4142}{0.1 \times |1 + 0.9j|} = b_0 \dfrac{1.4142}{0.1 \times 1.3454} = 10.512\, b_0.$

So $b_0 = 1/10.512 = \mathbf{0.0951}$, and

$$y(n) = 1.2728\, y(n-1) - 0.81\, y(n-2) + 0.0951\,[x(n) - x(n-2)].$$

(The true peak is at 0.252π, slightly above ω_0, with $|H| = 10.53\, b_0$. The question asks for normalisation at ω_0.)

(c) [1] With $N = 1$, $A(z^{-1}) = 1 - 0.8z$:

$$H_{ap}(z) = z^{-1}\frac{1 - 0.8z}{1 - 0.8z^{-1}} = \frac{-0.8 + z^{-1}}{1 - 0.8z^{-1}}.$$

The pole is at $z = 0.8$ and the zero at $z = 1/0.8 = 1.25$ (reciprocals). On the unit circle,

$$|-0.8 + e^{-j\omega}| = |e^{-j\omega}|\,|1 - 0.8e^{j\omega}| = |1 - 0.8e^{-j\omega}|$$

(a number and its conjugate have equal modulus). So $|H_{ap}(\omega)| = 1$.

(d) [1] $H_{ap}(\pi/2) = \dfrac{-0.8 - j}{1 + 0.8j}$. Then

$$\angle H_{ap}(\pi/2) = \angle(-0.8 - j) - \angle(1 + 0.8j) = -128.66^\circ - 38.66^\circ = \mathbf{-167.3}^\circ\ (-2.920\ \text{rad}).$$

Equivalently, $\theta(\omega) = -\omega - 2\arctan\dfrac{0.8\sin\omega}{1 - 0.8\cos\omega}$. At $\pi/2$ this is $-\pi/2 - 2\arctan 0.8 = -2.920$ rad.

(e) [1] For small ω, $\theta \approx -\omega - 2\dfrac{0.8\omega}{0.2} = -9\omega$. So $\tau(0) = \mathbf{9}$ samples. In general $\tau(0) = (1+a)/(1-a)$. An allpass filter changes phase (delay) without changing magnitude, so it is used as a **phase equaliser**: it corrects the nonlinear phase of another filter.

MARKS: (a) ½ for $H(z)$, ½ for the frequencies. (b) 1 for b_0, 1 for the difference equation. (c) 1. (d) 1. (e) ½ for the delay, ½ for the use.

COMMON MISTAKES

- Using $\omega_0 = 2\pi f_0$ without dividing by F_s.
- Writing $+2r\cos\omega_0$ in the denominator. The sign is negative.
- Leaving $b_0 = 1$, or guessing a value. Compute $|H(\omega_0)|$ directly from the factored form.
- Assuming the peak is exactly at ω_0. It is close (for r near 1) but not exact.
- Building the allpass with pole and zero both at 0.8 (that gives $H = 1$). The zero is at the **reciprocal** of the pole.

REREAD: D10 (resonators, notch, comb and allpass; L09 slides 17 and 23).

Question 9: Chebyshev type I order and ϵ [5]

(a) [1]

- $\omega_p = 2\pi(1)/10 = 0.2\pi$ and $\omega_s = 2\pi(2.5)/10 = 0.5\pi$.
- Prewarp with $T = 1$: $\Omega_p = 2\tan(0.1\pi) = \mathbf{0.6498}$ and $\Omega_s = 2\tan(0.25\pi) = \mathbf{2.0000}$.

(b) [1]

- $\delta_p = 10^{-0.5/20} = 0.9441$ and $\delta_s = 10^{-40/20} = 0.01$.
- $\epsilon^2 = 1/\delta_p^2 - 1 = 0.1220$, so $\epsilon = \mathbf{0.3493}$.

(c) [2] The lecture's formula uses inverse **hyperbolic** cosine, because both arguments are greater than 1:

$$N = \frac{\cosh^{-1}\left(\frac{1}{\epsilon}\frac{\sqrt{1-\delta_s^2}}{\delta_s}\right)}{\cosh^{-1}(\Omega_s/\Omega_p)} = \frac{\cosh^{-1}(286.26)}{\cosh^{-1}(3.0777)} = \frac{6.3501}{1.7898} = 3.548.$$

Round up: $\boxed{N = 4}$. (Check: $N = 4$ gives 47.0 dB at Ω_s, and $N = 3$ gives only 31.5 dB.) A Butterworth filter for the same specification needs $N = 5.03 \to 6$, which shows why Chebyshev is "more efficient".

(d) [1] Use $N = 5$: $T_5(3.0777) = \cosh(5 \times 1.7898) = 3850.5$. Equivalently, $T_5(x) = 16x^5 - 20x^3 + 5x$.

$$A_s = 10\log_{10}\left(1 + \epsilon^2 T_5^2\right) = 10\log_{10}\left(1 + 0.1220 \times 3850.5^2\right) = \mathbf{62.6\ dB}.$$

MARKS: (a) ½ each for the ω pair and the Ω pair. (b) 1. (c) 1 for correct substitution, 1 for N rounded up. (d) 1.

COMMON MISTAKES

- Typing $\cos^{-1}$ into the calculator. The argument is greater than 1, so you get an error or a complex number. The slide's "$\cos^{-1}$" means $\cosh^{-1}$ here, as the T9 Q7 solution confirms.
- Not prewarping (see Q11(c)).
- Rounding N down.
- Using $\delta_s = 10^{-40/10}$. Amplitude specifications use $/20$.

REREAD: D11 (prewarping, bilinear design steps), D12 (Chebyshev formulas; T9 Q7 is the type II version).

Question 10: Completing a linear-phase FIR filter [5]

(a) [1] $M = 6$. The end samples are $h(0) = -1$ and $h(5) = 1$, which are negatives of each other, so the filter must be **antisymmetric**: $h(n) = -h(5-n)$. This is consistent with $h(1) = 2 = -h(4)$. Therefore

$$h(3) = -h(2) = \mathbf{-4}, \qquad h(n) = \{\underset{\uparrow}{-1}, 2, 4, -4, -2, 1\}.$$

(With M even there is no middle sample, so it is not forced to zero as in T9 Q8.)

(b) [2] Factor out the centre $e^{-j2.5\omega}$ and pair the samples:

$$H(\omega) = e^{-j2.5\omega}\left[-\left(e^{j2.5\omega} - e^{-j2.5\omega}\right) + 2\left(e^{j1.5\omega} - e^{-j1.5\omega}\right) + 4\left(e^{j0.5\omega} - e^{-j0.5\omega}\right)\right]$$

$$\boxed{H(\omega) = \underbrace{2\left[4\sin\tfrac{\omega}{2} + 2\sin\tfrac{3\omega}{2} - \sin\tfrac{5\omega}{2}\right]}_{A(\omega)} e^{-j(2.5\omega - \pi/2)}}$$

(c) [1] $A(\omega) \geq 0$ for $0 \leq \omega \leq \pi$ (checked numerically: it never changes sign). So

$$\theta(\omega) = -2.5\omega + \frac{\pi}{2}, \quad 0 < \omega \leq \pi,$$

which is linear, with group delay $\tau = (M-1)/2 = \mathbf{2.5}$ samples. (Spot check: $H(\pi/2) = -7 - 7j$, with angle $-3\pi/4 = -2.5(\pi/2) + \pi/2$.)

(d) [1]

- $H(0) = \sum h(n) = -1 + 2 + 4 - 4 - 2 + 1 = \mathbf{0}$
- $H(\pi) = \sum (-1)^n h(n) = -1 - 2 + 4 + 4 - 2 - 1 = \mathbf{2}$

An antisymmetric FIR filter always has $H(0) = 0$ (a zero at $z = 1$), so it **cannot** be a lowpass filter. This one passes high frequencies (highpass or bandpass behaviour).

MARKS: (a) ½ for antisymmetric with a reason, ½ for the value. (b) 1 for factoring out $e^{-j2.5\omega}$, 1 for $A(\omega)$ and the $\pi/2$. (c) 1. (d) ½ for the values, ½ for the conclusion.

COMMON MISTAKES

- Choosing symmetric because "most filters are". Check the first and last samples.
- Writing $h(3) = 0$ by copying T9 Q8. That rule is only for odd M.
- Losing the $+\pi/2$. It comes from $e^{jx} - e^{-jx} = 2j\sin x$ and $j = e^{j\pi/2}$.
- Using a group delay of $M/2 = 3$. It is $(M-1)/2$.

REREAD: D12 (linear-phase FIR, symmetric and antisymmetric cases).

Question 11: Short answers [4]

(a) [1] The ideal LPF has $h(n) = \dfrac{\sin \omega_c n}{\pi n}$ (T9 Q1). It is **noncausal** ($h(n) \neq 0$ for $n < 0$), and it is of **infinite duration**, so it cannot be computed with finite memory and operations. (It is also not absolutely summable.) ½ mark per reason.

(b) [2]

- Bin spacing: $F_s/N = 8000/1024 = \mathbf{7.81\ Hz}$.
- Resolution is set by the record length, not by N: $1/T_0 = F_s/L = 8000/64 = \mathbf{125\ Hz}$. This is greater than the 50 Hz spacing, so the tones **cannot be resolved**. Zero padding only interpolates the same smeared spectrum (a simulation shows one peak near 1026 Hz).
- Minimum record: $L \geq F_s/\Delta F = 8000/50 = \mathbf{160}$ samples, so $T_0 = 160/8000 = \mathbf{20\ ms}$.

Marks: ½ for the bin spacing, 1 for "no" with a reason, ½ for L and T_0.

(c) [1] The bilinear map gives $\omega = 2\arctan(\Omega T/2) = 2\arctan(\pi/4) = 1.3316$ rad $= 0.4239\pi$. That corresponds to $f = 0.4239 \times 5000 = \mathbf{2119\ Hz}$, not 2500 Hz. The map $\Omega \to \omega$ is nonlinear (frequency warping): the whole $j\Omega$ axis is squeezed onto $-\pi < \omega < \pi$, and $\Omega = \infty$ maps to $\omega = \pi$. That is why the critical edges must be prewarped with $\Omega = \frac{2}{T}\tan\frac{\omega}{2}$.

COMMON MISTAKES

- Saying zero padding "improves resolution". It improves how finely you sample the DTFT, not your ability to separate tones. The lecture's wording ("better resolution for sampled DTFT") means the finer sampling only.
- Saying the ideal LPF is unstable or has infinite gain. The course's two reasons are causality and infinite length.

REREAD: D6 (frequency analysis with the DFT), D11 (warping), D12 (ideal LPF and windows).

Score table

Score	What it means two days out
50–60	Ready. Use tonight for the checklist only.
40–49	Solid. Redo the one or two questions you dropped most marks on, open book, tomorrow.
30–39	Reread the packs flagged on your weakest three questions. Redo those questions without looking.
below 30	Focus on method, not speed: work through D7–D9 and D11–D12 worked examples, then redo Q6, Q7 and Q9.

Final-evening checklist (for the night before the exam)

Do this in about 45 minutes. Do not start new material.

Pack your bag

- ☐ Printed lecture slides L02–L10, all tutorial solutions T1–T9, your Test 1 Reference, the L05–L10 reference sheet (REFD), and both mock marking guides.
- ☐ A calculator that does complex numbers and `cosh⁻¹`, plus a spare battery. Check it is in **radians** mode.
- ☐ Pens, pencil and ruler (for stem plots, pole-zero plots and sketches).

Tab these pages so you can find them in 10 seconds

- ☐ DTFT property table (L05) and Parseval.
- ☐ DFT properties and overlap-save/overlap-add slides (L06, slides 20–27).
- ☐ z-transform pairs and ROC rules (L07), the unilateral shift formulas (L08 slide 13).
- ☐ Resonator, notch, comb and allpass formulas (L09 slides 17–25).
- ☐ Butterworth and Chebyshev N formulas, prewarp formula, window table (L10 slides 7, 17, 24, 41).

Ten things to say out loud once

1. Correlation peak at lag d gives the delay. Range $r = dc/(2F_s)$.
2. Complex modes $\rho e^{\pm j\theta}$ give $\rho^n[A\cos\theta n + B\sin\theta n]$. Fit A and B from $h(0)$ and $h(1)$.
3. $X(0) = \sum x$, $X(\pi) = \sum(-1)^n x$, $\int X e^{j\omega n_0} = 2\pi x(n_0)$, $\int |X|^2 = 2\pi \sum |x|^2$. Find the origin first.
4. DFT convolution is linear only if $N \geq L + M - 1$. Deconvolution fails where $H(k) = 0$.
5. Overlap-save: overlap the inputs by $M - 1$ and throw away the first $M - 1$ outputs. Overlap-add: split the inputs and add the tails.

6. n pole radii give $n+1$ ROCs. A pole inside the ROC is causal, $p^n u(n)$. A pole outside is anticausal, $-p^n u(-n-1)$. The unit circle in the ROC means a DTFT exists (stable).
7. Unilateral shift: $y(n-1) \to z^{-1}Y^+ + y(-1)$ and $y(n-2) \to z^{-2}Y^+ + z^{-1}y(-1) + y(-2)$.
8. Resonator: $\omega_0 = 2\pi f_0/F_s$, denominator $1 - 2r\cos\omega_0 z^{-1} + r^2 z^{-2}$. Allpass zeros are the reciprocals of its poles.
9. Prewarp with $\Omega = 2\tan(\omega/2)$ ($T=1$). Amplitude in dB uses $/20$. Chebyshev N uses $\cosh^{-1}$. Always round N **up**.
10. Linear-phase FIR: compare the end samples to choose symmetric or antisymmetric. Odd-length antisymmetric means the middle sample is 0. Group delay is $(M-1)/2$. Antisymmetric means $H(0) = 0$.

SANITY CHECKS TO USE IN THE EXAM

- ☐ After any partial fraction expansion, put one value of z back in, or check $x(0)$ by long division or recursion.
- ☐ After any LCCDE answer, check $y(0)$ and $y(1)$ by recursion (costs 30 seconds).
- ☐ After any DFT, check that $X(0) = \sum x(n)$ and that $X(N-k) = \overline{X(k)}$ for real x.
- ☐ After any filter design, check that $|H|$ meets the specification at both edges.

EXAM TECHNIQUE

- ☐ Read the whole paper first (5 minutes). Do the questions you are surest of first.
- ☐ If you are stuck on a part, use the “assuming the answer is…” value and move on. Never leave a later part blank.
- ☐ Write the formula before the numbers. The method marks are there even if the arithmetic slips.
- ☐ Sleep. A rested 2 hours beats one more hour of revision tonight.